Probability 1, Autumn 2016, Problem sheet 1

To be discussed on the week 3 Oct... 7 Oct.
Problems marked with "PrCl" are discussed in the large problem class on Friday 30 Sep.
Mandatory HW's are marked with "**HW**", they are due on the week 10 Oct... 14 Oct.[1]
Solutions will be available on Blackboard on the 15th Oct.

1.1 How many $\{1, 2, \ldots, n\} \to \{0, 1\}$ functions are there? (That is, functions that assign either 0 or 1 to integers between 1 and n.) And how many $\{0, 1\} \to \{1, 2, \ldots, n\}$ functions?

PrCl **1.2** I have 4 Maths, 3 Chemistry, 2 History and one French language book (no two are identical). In how many ways can I order them on my bookshelf if I want books of the same topics to stay next to each other?

HW 1.3 (a) In how many ways can three girls and three boys sit in a row? (Notice: people are never considered indistinguishable.)

(b) In how many ways can three girls and three boys sit in a row if girls sit next to each other and boys sit next to each other?

(c) In how many ways can three girls and three boys sit in a row if only boys are required to sit next to each other?

(d) In how many ways can three girls and three boys sit in a row if no two children of the same sex are allowed to sit next to each other?

PrCl **1.4** Out of 5 women and 7 men we wish to form a committee of 2 women and 3 men.

(a) In how many ways can we do that?

(b) And how many ways if two of the 7 men refuse to serve together?

HW 1.5 Out of 8 women and 6 men we wish to form a committee of 3 women and 3 men.

(a) In how many ways can we do that?

(b) How many ways if two of the 6 men refuse to serve together?

(c) How many ways if two of the 8 women refuse to serve together?

(d) How many ways if one of the women and one of the men refuse to serve together?

1.6 12 ladies and 13 gentlemen attend a dance class. In how many ways can the instructor select 6 ladies, 6 gentlemen, and then pair them?

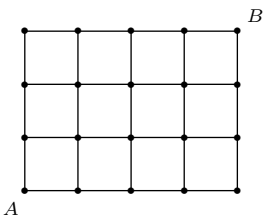
1.7 A walker starts from 0, and can make a jump of length one in each step in either directions on the integer line $\mathbb{Z}$. She makes 10 steps in total and then stops.

(a) How many different trajectories of her motion are possible?

(b) In how many positions can she end up after 10 steps?

(c) Suppose she ends up at position 4. In how many ways could she get there in 10 steps?

HW 1.8 On the graph below, how many ways are there from A to B only using steps of length one in the Northern or Eastern direction?



1.9 10 children want to play football.

(a) How many ways are there to form teams A and B of sizes 5 each?

(b) How many ways are there to form two teams of sizes 5 each? (!)

[1] Details of how to hand in are to be discussed with your tutor.

HW 1.10 Out of n people we want to form a committee of k, one of whom will be the president of the committee.

(a) By picking the committee first, then its president, argue that there are $\binom{n}{k} k$ ways to do this.

(b) By choosing the non-president members of the committee first, then picking the president from the remaining people, argue that there are $\binom{n}{k-1}(n-k+1)$ many ways.

(c) Pick the president first, then select the non-president members, and argue that the number of ways is $n\binom{n-1}{k-1}$.

(d) Draw the conclusion:

$$\binom{n}{k} k = \binom{n}{k-1}(n-k+1) = n\binom{n-1}{k-1}.$$

Prove this also analytically (by writing out all the factorials).